

Problem 8.1

The amount of interest earned on A for one year is \$336, while the equivalent amount of discount is \$300. Find A .

Solution

Let i be the effective rate of interest and d the effective rate of discount for one year.

Since the amount of interest earned on A for one year is \$336,

$$Ai = 336,$$

so

$$i = \frac{336}{A}.$$

Since the equivalent amount of discount is \$300,

$$Ad = 300,$$

so

$$d = \frac{300}{A}.$$

Using the relationship between the effective discount rate and the effective interest rate,

$$d = \frac{i}{1 + i}.$$

Substitute the expressions for d and i :

$$\frac{300}{A} = \frac{\frac{336}{A}}{1 + \frac{336}{A}}.$$

Multiply both sides by A :

$$300 = \frac{336}{1 + \frac{336}{A}}.$$

Now multiply both sides by $1 + \frac{336}{A}$:

$$300 \left(1 + \frac{336}{A} \right) = 336.$$

Expanding,

$$300 + \frac{300 \cdot 336}{A} = 336.$$

Thus,

$$\frac{300 \cdot 336}{A} = 36.$$

So,

$$A = \frac{300 \cdot 336}{36} = 2800.$$

Therefore,

$$\boxed{A = 2800}.$$